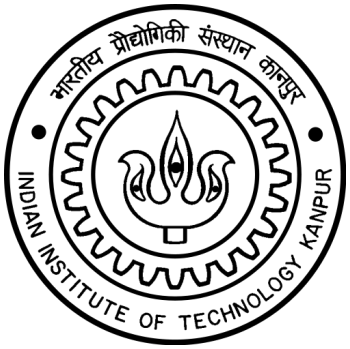


SPA617

RADIO ASTRONOMY LABORATORY

RF Components : Low Pass Filter, Notch Filter, Mixer, Power Splitter and Combiner

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1. Introduction

1.1. My Understanding about RF components

Before performing this experiment, my understanding of RF components was primarily that they are specialized components designed to operate on and manipulate signals in the radio-frequency range, much like coaxial cables are specialized transmission lines used to guide RF signals with controlled impedance and reduced signal loss. Although the concepts of low-pass and notch filters were already familiar to me, the practical operating characteristics and physical implementation of these components were relatively new. In particular, the construction and working of the quarter-wave open-circuit stub as a notch filter was an interesting concept, as the transformation of an open circuit into a low impedance at the quarter-wavelength condition provided a practical demonstration of how transmission-line behaviour can be exploited to reject a specific frequency.

Another important aspect of the experiment was understanding that RF components are not ideal and that losses play a significant role in their practical behaviour. Measuring quantities such as insertion loss, attenuation, conversion loss, and power-division loss helped me understand how the response of an RF component differs from its ideal theoretical behaviour. These losses are important when designing an RF signal chain because even passive components can significantly affect the power available at different stages of the system.

The most interesting part of the experiment for me was the study of the frequency components produced by the mixer. Unlike a linear component, a mixer is a nonlinear device, and the interaction of the RF and local-oscillator signals generates new frequency components. The fundamental mixing process produces the sum and difference frequencies, $(f_{\mathrm{RF}} + f_{\mathrm{LO}})$ and $(|f_{\mathrm{RF}} - f_{\mathrm{LO}}|)$. Due to the nonlinear nature of the mixer, higher-order terms can also be generated, giving rise to harmonics and additional mixing products. This demonstrated to me how nonlinear RF devices can be used deliberately for frequency conversion, while also producing unwanted spectral components that may need to be filtered out in a practical receiver system.

Overall, the experiment helped me move from understanding RF components primarily as specialized hardware to understanding how their physical construction, transmission-line behaviour, losses, and nonlinear characteristics determine their actual response to RF signals.

2. Aim

To experimentally characterize and study the operation of different RF components used in an RF receiver chain, namely a low-pass filter, a notch filter, an RF mixer, a power splitter, and a power combiner.

3. Objectives of the Experiment

The objectives of the experiment were:

1. To characterize the transmission response of a low-pass RF filter and determine its cutoff frequency and insertion loss.
2. To demonstrate the operation of a quarter-wave open-circuit stub as a notch filter and determine its notch frequency.
3. To observe frequency conversion using an RF mixer and determine the power of the respective frequency components.
4. To characterize a two-way power splitter by measuring the power delivered at its output ports.

5. To study the operation of a power combiner by observing the frequency components at its combined output.

4. Apparatus Required

The following equipment and components were used during the experiment:

- RF signal generator(s)
- Spectrum analyser
- Digital storage oscilloscope
- Low-pass RF filter
- RF Tee connector
- Coaxial cable
- RF mixer
- Two-way power splitter/combiner
- 50 Ω termination

5. Theory

5.1. Low-Pass Filter

An RF filter allows selected frequency components of a signal to pass while attenuating unwanted frequency components. A low-pass filter passes frequencies below its cutoff frequency and increasingly attenuates frequencies above the cutoff.

The transmission performance of the filter can be characterized using its insertion loss. If P_{in} is the power measured without the filter and P_{out} is the power measured after inserting the filter, the insertion loss is

$$IL = P_{\text{in}} - P_{\text{out}} \tag{1}$$

where the powers are expressed in dBm.

A smaller insertion loss indicates better transmission through the filter.

5.2. Notch Filter Using a Quarter-Wave Stub

A notch filter, also called a band-reject filter, attenuates a narrow range of frequencies while allowing frequencies outside the stopband to pass.

In this experiment, the notch filter was implemented using an open-ended coaxial stub connected through an RF Tee connector. At the quarter-wave resonant frequency, the open circuit at the far end of the stub is transformed approximately into a low impedance at the Tee junction. Consequently, the signal at this frequency is strongly attenuated, producing a notch in the transmission response.

The approximate physical length-frequency relationship is

$$L = \frac{\lambda}{4} \tag{2}$$

or

$$f_0 = \frac{VF\,c}{4L} \tag{3}$$

where L is the physical length of the stub, VF is its velocity factor, c is the speed of light in vacuum, and f_0 is the expected notch frequency.

The measured notch frequency may differ slightly from the calculated value because of connector dimensions, cable effects, and the open-end effect.

5.3. RF Mixer

An RF mixer is a nonlinear three-port device having RF, local oscillator (LO), and intermediate-frequency (IF) ports. When signals having frequencies f_{RF} and f_{LO} are applied to the RF and LO ports, respectively, the output contains frequency components corresponding to the sum and difference of the two input frequencies.

Thus,

$$f_{\mathrm{IF}} = f_{\mathrm{RF}} + f_{\mathrm{LO}} \tag{4}$$

and

$$f_{\mathrm{IF}} = \left| f_{\mathrm{RF}} - f_{\mathrm{LO}} \right|. \tag{5}$$

The desired component can subsequently be selected using a filter.

The conversion loss of the mixer is defined as

$$\text{Conversion Loss} = P_{\mathrm{RF}} - P_{\mathrm{IF}} \tag{6}$$

where P_{RF} is the RF input power and P_{IF} is the measured power of the selected IF component. [**?**]

5.4. Power Splitter

A power splitter divides the power supplied at its input among two or more output ports. For an ideal two-way equal power splitter, the power at each output is 3 dB below the input power.

For an input power P_{in}, the losses at the two output ports are

$$L_1 = P_{\mathrm{in}} - P_{\mathrm{out1}} \tag{7}$$

and

$$L_2 = P_{\mathrm{in}} - P_{\mathrm{out2}}. \tag{8}$$

For an ideal splitter,

$$L_1 = L_2 = 3\,\mathrm{dB}. \tag{9}$$

In practice, the measured losses can be slightly greater than 3 dB because of the insertion loss and non-ideal characteristics of the component.

5.5. Power Combiner

A power combiner performs the reverse operation of a power splitter by combining signals applied at multiple input ports into a common output.

The output spectrum depends on the frequencies, amplitudes, and relative phases of the input signals. Therefore, when two signals are applied at different frequencies, both frequency components can be observed at the combined output. When signals of the same frequency are combined, their resulting amplitude depends on their relative phase. To remove the phase dependence, we construct two coherent sources by first connecting a power splitter to the same power source and then using the two outputs of the power splitter as the inputs for the power combiner.

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6. Experimental Setup

The RF components were characterized using RF signal generator(s), a spectrum analyser, RF cables, and appropriate connectors and terminations. The experimental setups used for the individual components are described below.

6.1. Low-Pass Filter

For the low-pass filter measurement, the RF signal generator was connected to the input of the low-pass filter. The output of the filter was connected to the spectrum analyser.

The setup was therefore:

$$\text{Signal Generator} \longrightarrow \text{Low-Pass Filter} \longrightarrow \text{Spectrum Analyser}.$$

The frequency of the signal generator was varied over the required range, and the corresponding output power was measured using the spectrum analyser. The measured response was used to characterize the frequency-dependent attenuation of the filter and determine its cutoff frequency.

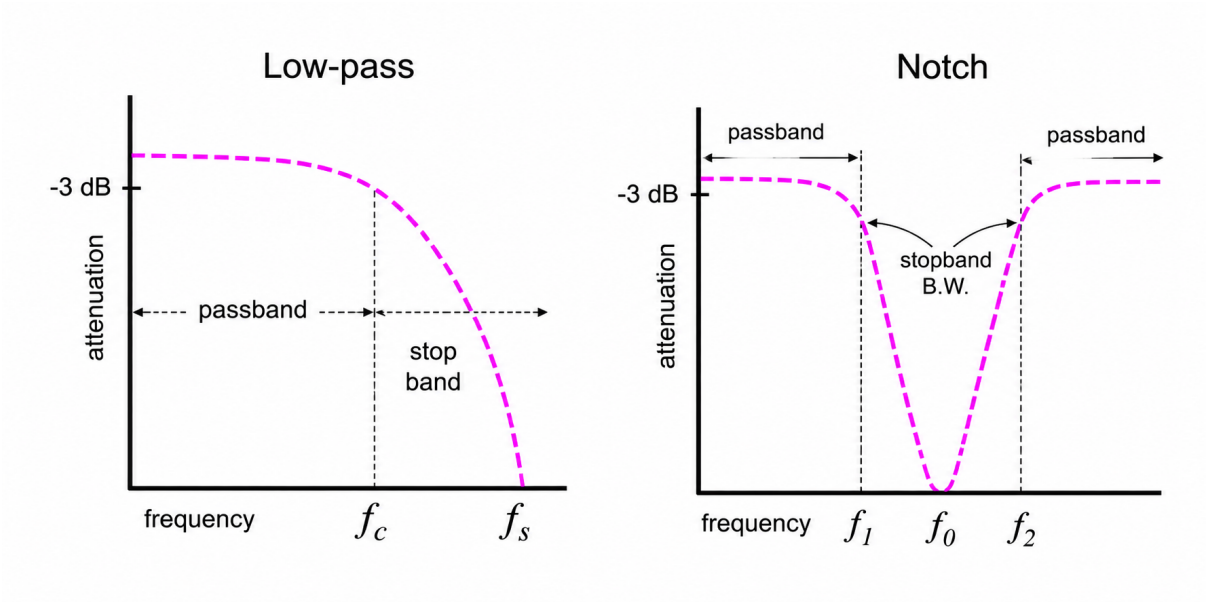


Figure 1: Typical frequency responses of low-pass and notch filters, showing the passband, stopband, and characteristic frequencies.

6.2. Notch Filter

The notch filter was implemented using an open-ended coaxial stub connected to an RF Tee connector. The signal generator was connected to one port of the Tee, while the spectrum analyser was connected to another port. The open-ended coaxial stub was connected to the remaining port of the Tee.

The setup was:

Open-ended coaxial stub
|
Signal Generator ⟶ [Tee Connector] ⟶ Spectrum Analyser.

The output power was measured over a range of frequencies around the expected notch frequency. The response obtained with the stub was compared with the reference response without the stub. The frequency at which maximum attenuation occurred was identified as the notch frequency.

6.3. Mixer

For the mixer characterization, two signal generators were connected to the RF and LO ports of the mixer. Both input signals were applied at the specified power level of 0 dBm. The IF port of the mixer was connected to the spectrum analyser.

The setup was:

Signal Generator (RF) ⟶ [Mixer] ⟶ Spectrum Analyser
Signal Generator (LO) ⟶

The RF and LO frequencies were varied as required, and the output spectrum was observed using the spectrum analyser. The sum and difference frequency components generated by the mixer were identified and compared with their theoretical values.

6.4. Power Splitter

For the power splitter characterization, the output of the signal generator was connected to the input port of the two-way power splitter. One output port of the splitter was connected to the spectrum analyser, while the other output port was connected to a suitable termination.

The setup was:

Signal Generator ⟶ [Power Splitter] ⟶ { Spectrum Analyser, Termination.

The output ports were interchanged so that the power at both output ports could be measured. The measured output powers were compared with the ideal 3 dB power division of an equal two-way splitter.

6.5. Power Combiner

For the power combiner characterization, the signal generator was first connected to the input of the power splitter. The two output ports of the splitter were then connected to the two input ports of the power combiner. The output of the combiner was connected to the spectrum analyser.

The setup was:

Signal Generator ⟶ Power Splitter ⟶ Power Combiner ⟶ Spectrum Analyser.

The splitter divided the signal into two paths, which were subsequently supplied to the two input ports of the combiner. The combined output was observed using the spectrum analyser. This arrangement allowed the behaviour of the signal after splitting and subsequent recombination to be examined.

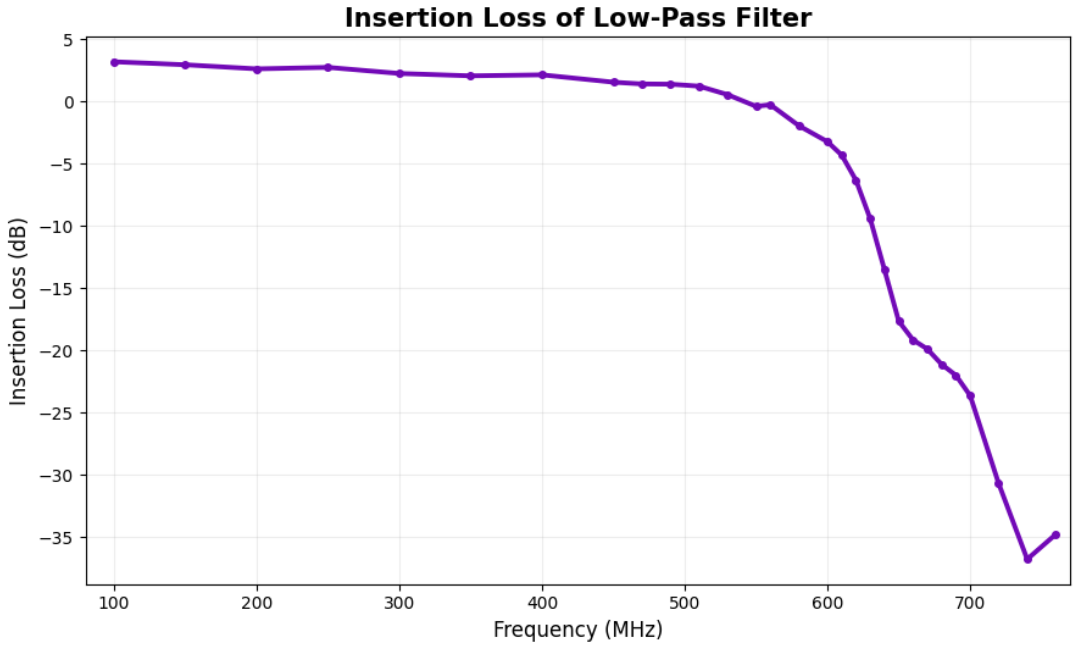


Figure 2: Measured insertion loss of the low-pass filter as a function of frequency.

7. Observations and Plots

7.1. Transmission response and insertion loss

The output power of the low-pass filter was measured over a range of frequencies, both with and without the filter. The cable-only measurements were used to account for the attenuation introduced by the connecting cables, and the resulting net power was calculated to obtain the frequency response of the filter. The cutoff frequency for the low pass filter was labelled as **400 MHz**.

Table 1: Measured power response and insertion loss of the low-pass filter.

Frequency (MHz)	Power with filter (dBm)	Power without filter (dBm)	Net power (dB)	Insertion loss (dB)
100	-5.49	-5.70	0.21	3.21
150	-5.73	-5.70	-0.03	2.97
200	-6.06	-5.70	-0.36	2.64
250	-5.94	-5.70	-0.24	2.76
300	-6.43	-5.70	-0.73	2.27
350	-6.62	-5.70	-0.92	2.08
400	-6.81	-5.97	-0.84	2.16
450	-7.40	-5.97	-1.43	1.57
470	-7.54	-5.97	-1.57	1.43
490	-7.56	-5.97	-1.59	1.41
510	-7.78	-6.03	-1.75	1.25
530	-8.46	-6.03	-2.43	0.57
550	-9.40	-6.03	-3.37	-0.37
560	-9.27	-6.03	-3.24	-0.24
580	-10.96	-6.03	-4.93	-1.93
600	-12.24	-6.03	-6.21	-3.21
610	-13.40	-6.11	-7.29	-4.29
620	-15.42	-6.11	-9.31	-6.31
630	-18.52	-6.11	-12.41	-9.41
640	-22.62	-6.11	-16.51	-13.51
650	-26.75	-6.11	-20.64	-17.64
660	-28.49	-6.35	-22.14	-19.14
670	-29.22	-6.35	-22.87	-19.87
680	-30.45	-6.35	-24.10	-21.10
690	-31.33	-6.35	-24.98	-21.98
700	-32.96	-6.35	-26.61	-23.61
720	-39.78	-6.10	-33.68	-30.68
740	-45.86	-6.10	-39.76	-36.76
760	-43.86	-6.10	-37.76	-34.76

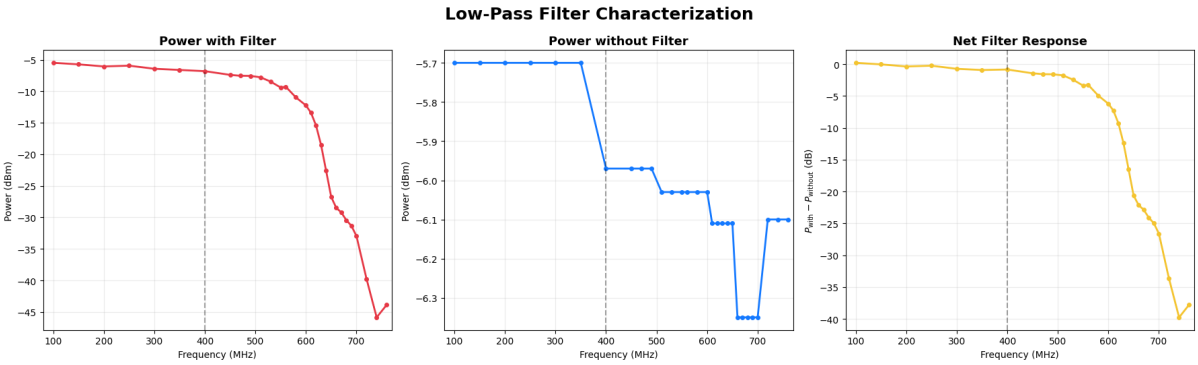


Figure 3: Measured power response of the low-pass filter with and without the filter, along with the corresponding net power response.

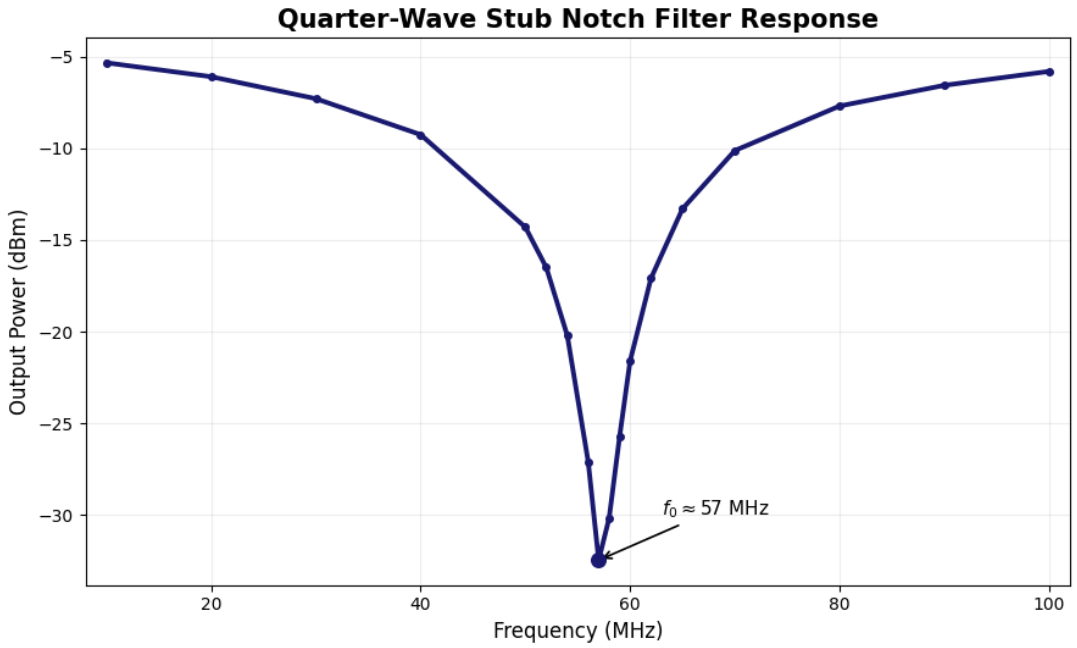


Figure 4: Measured frequency response of the quarter-wave stub notch filter. The minimum output power occurs at approximately 57 MHz.

7.2. Quarter-wave open-circuit stub notch filter

The output power of the quarter-wave stub notch filter was measured over a range of frequencies around the expected resonant frequency. The measured response shows a reduction in output power near the notch frequency, corresponding to the rejection produced by the open-ended coaxial stub.

Table 2: Measured output power of the quarter-wave stub notch filter.

Frequency (MHz)	Output Power (dBm)
10	-5.33
20	-6.09
30	-7.29
40	-9.25
50	-14.30
52	-16.50
54	-20.20
56	-27.09
57	-32.47
58	-30.17
59	-25.72
60	-21.60
62	-17.08
65	-13.30
70	-10.12
80	-7.69
90	-6.56
100	-5.80

7.3. RF Mixer

The output spectrum of the mixer was observed for different RF input frequencies while maintaining the local oscillator frequency at 30 MHz. The peak powers of the first harmoics for the difference-frequency and sum-frequency components were recorded from the spectrum analyser.

Table 3: Observed output components of the RF mixer.

f_{RF} (MHz)	$f_{RF} - f_{LO}$ (MHz)	Peak IF_1 (dBm)	$f_{RF} + f_{LO}$ (MHz)	Peak IF_2 (dBm)
120	90	-12.32	150	-12.48
150	120	-12.56	180	-12.50
180	150	-12.40	210	-12.51

The local oscillator frequency was maintained at $f_{LO} = 30$ MHz. The two observed output components correspond to the difference and sum frequencies given by

$$f_{IF_1} = |f_{RF} - f_{LO}|$$

and

$$f_{IF_2} = f_{RF} + f_{LO}.$$

7.4. Power Splitter

The output powers at the two ports of the power splitter were measured for different input frequencies while maintaining the input power at 0 dBm. The two output ports were measured separately by interchanging the spectrum analyser and termination connections.

Table 4: Measured output powers of the RF power splitter.

Frequency (MHz)	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)
150	0	-6.32	-6.19
200	0	-5.85	-5.89
300	0	-6.58	-6.59

7.5. Power Combiner

The output power of the power combiner was measured at different frequencies after the signal was divided and subsequently recombined. The resulting output power was recorded using the spectrum analyser to characterize the response of the combiner. The input power provided to the power splitter was 0 dBm, which would have resulted in splitting of the output power to -3 dBm at each terminal. This would have resulted in the overall output of 0 dBm at the output terminal of the power combiner.

Table 5: Measured output power of the RF power combiner.

Frequency (MHz)	Output Power (dBm)
50	-2.89
100	-3.20
150	-3.12
200	-3.68

8. Results and Discussion

8.1. Low-Pass Filter

The insertion loss of the low-pass filter was calculated from the measured power values and plotted as a function of frequency. The measured response does not correspond exactly to the nominal cutoff frequency of approximately 400 MHz specified for the filter. The deviation can be attributed to non-idealities in the experimental setup, including attenuation and losses introduced by the cables, connectors, and other components in the measurement system. These additional losses can affect the measured response and result in a deviation from the nominal filter characteristics.

Nevertheless, the overall frequency response follows the expected low-pass behaviour. The output power remains relatively high at lower frequencies and decreases significantly as the frequency is increased beyond the passband. Thus, the experimental results successfully demonstrate the frequency-selective behaviour of the low-pass filter.

8.2. Notch Filter

The notch filter produced a clear rejection region around its resonant frequency. The measured output power decreases sharply within a narrow band of frequencies, demonstrating that the quarter-wave coaxial stub effectively suppresses signals within this frequency range while allowing frequencies outside the rejection band to pass. The minimum measured output power occurred at approximately 57 MHz, confirming the expected notch-filter behaviour.

8.3. RF Mixer

The mixer characterization produced the expected frequency components at the output. The observed components occurred at the sum and difference frequencies of the applied RF and local oscillator signals, in agreement with the theoretical relations

$$f_{\mathrm{IF}_1} = |f_{\mathrm{RF}} - f_{\mathrm{LO}}|$$

and

$$f_{\mathrm{IF}_2} = f_{\mathrm{RF}} + f_{\mathrm{LO}}.$$

The experiment also demonstrated the generation of additional higher-order spectral components arising from the nonlinear operation of the mixer. The observed components appeared at the expected frequencies, with their measured powers being of comparable magnitude. This demonstrates the frequency-mixing and nonlinear characteristics of the RF mixer.

8.4. Power Splitter

For an ideal two-way power splitter with an input power of 0 dBm, the power at each output port is expected to be approximately -3 dBm, corresponding to an ideal 3 dB power division. In the experiment, the measured output powers were approximately -6 dBm at the two output ports.

The additional attenuation can be attributed to losses associated with the splitter itself, as well as losses introduced by the connecting cables, connectors, and measurement setup. Thus, although the measured values differ from the ideal value, the two output ports exhibit approximately equal power division, as expected for an equal power splitter.

8.5. Power Combiner

For an ideal power combiner, the signals applied at the input ports are combined at the output, with the resulting output power depending on the input powers and their relative phases. The measured output powers in the experiment were approximately in the range of -2.5 to $-3.5\,\mathrm{dBm}$, rather than the ideal value expected from lossless combining.

The measured losses are comparable in magnitude to those observed during the splitter characterization. This indicates that the practical splitter/combiner component and the associated measurement system introduce a relatively consistent amount of attenuation over the measured frequency range. The results therefore demonstrate the practical non-ideal behaviour of RF power combining and splitting components.

AI Declaration

AI tools (primarily ChatGPT) were used as an assistive tool during the preparation of this report. The following uses of AI are acknowledged:

1. **Content refinement:** ChatGPT was used to enhance the clarity, structure, grammar, and overall quality of the content written in the report. The experimental observations, interpretations, and underlying ideas presented in the report were provided by me.
2. **Theoretical figures:** AI-based image generation was used to create the ideal theoretical frequency-response figure for the low-pass and notch filters shown in Figure 1. The experimental plots, however, were generated using Python from the experimentally measured data.
3. **LaTeX formatting:** ChatGPT was used to assist with the formatting of the report in LaTeX, including the preparation and formatting of tables, figures, equations, and other report elements.
4. **Conceptual understanding:** ChatGPT was used as a learning aid to improve my understanding of selected RF concepts, particularly the operating principle of the coaxial quarter-wave stub used in the notch-filter experiment.

Declaration by Akshar Kishore